BMC Cancer

STUDY PROTOCOL

Open Access

[^{18}F]F-FAPI PET/CT and LAparoscopy in STagIng advanced gastric Cancer: a multicenter prospective study (PLASTIC-3 study)

Lianne Triemstra[1,2†], Sarah W. J. M. Spruijt[2†], Dennis Vriens[3], Erik Vegt[4], Wilma E. Mesker[5], Lukas J. A. C. Hawinkels[6], Sopany Saing[7], Marije Slingerland[8], Vera Kemp[9], Stijn A. L. P. Crobach[10], Jelle J. Goeman[11], Martin Pool[12], Petra Dibbets-Schneider[2], Roel J. Bennink[2], Mark I. van Berge Henegouwen[13,14], Marc J. van Det[15], Boudewijn J. van Etten[16], Henk H. Hartgrink[5], Richard van Hillegersberg[1], Marnix G. E. H. Lam[17], Nanda M. Smulders[18], Misha D. P. Luyer[19], Walter Noordzij[20], Emilia C. Owers[21], Wim J. G. Oyen[22,23], Johanna W. van Sandick[24], Wendy Schreurs[25], Wobbe O. de Steur[5], Jan H. M. B. Stoot[26], Leo E. Weijs[27], Bas P. L. Wijnhoven[28], Bart P. L. Witteman[29], Jelle P. Ruurda[1], Lioe-Fee de Geus-Oei[2,30*] and PLASTIC-3 Study Group

Abstract

Background Treatment decisions for locally advanced gastric adenocarcinoma rely on staging with gastroscopy, computed tomography (CT), and staging laparoscopy (SL). However, CT often fails to detect peritoneal and other distant metastases, and SL is an invasive procedure with complication risks and costs. Therefore, an accurate non-invasive imaging biomarker to detect such metastases could improve the care for gastric cancer patients. Radiolabeled fibroblast-activating protein inhibitors (FAPI), targeting stromal cancer-associated fibroblasts, are promising radiotracers. Preliminary Asian studies indicate that FAPI PET/CT outperforms [^{18}F]FDG PET/CT in tracer uptake, tumor-to-background ratio and diagnostic sensitivity in gastric cancer. However, larger studies using FAPI PET/CT in homogeneous, representative cohorts are needed to validate and translate these findings to Western populations. This study evaluates the clinical impact and diagnostic accuracy of [^{18}F]AlF-FAPI-74 PET/CT for optimizing patient staging, aiming to reduce futile toxic neoadjuvant and surgical treatments. This may help optimize treatment pathways, enhance quality of life (QoL), and reduce patient burden and healthcare costs.

Methods This Dutch multicenter prospective clinical trial will recruit 250 patients with resectable, locally advanced gastric adenocarcinoma (cT3-4N0-3M0) as determined by gastroscopy and CT, planned for curative treatment. The

† Lianne Triemstra and Sarah W. J. M. Spruijt contributed equally to this work.

*Correspondence:
Lioe-Fee de Geus-Oei
l.oei@utwente.nl

Full list of author information is available at the end of the article

modality under investigation is [^{18}F]AlF-FAPI-74 PET/CT. The primary outcome is the proportion of patients in whom [^{18}F]AlF-FAPI-74 PET/CT leads to a change in treatment intent. Secondary outcomes include diagnostic performance, patient burden from incidental or false-positive PET/CT findings, diagnostic delay, and clinicopathological correlation between [^{18}F]AlF-FAPI-74 PET/CT and intra-operative findings during SL. The composite reference standard for 'true' (intraperitoneal) tumor burden includes SL-based peritoneal cancer index scores, biopsy, imaging and intra-operative findings. If [^{18}F]AlF-FAPI-74 PET/CT proves reliable, its cost-effectiveness will be compared to the PLASTIC-study. Recruitment started July 3rd 2025 and will continue for 36 months.

Discussion This study hypothesizes that [^{18}F]AlF-FAPI-74 PET/CT can detect two-thirds of all gastric cancer patients with M1-disease according to the reference standard. Additionally, an [^{18}F]AlF-FAPI-74 PET/CT scan may reduce healthcare costs, optimize treatment pathways and patients' QoL. The study findings will be relevant for countries with similar treatment algorithms and healthcare systems.

Trial registration ClinicalTrials.gov, NCT07018661. This trial was registered prospectively on June 12th, 2025.

Keywords Gastric cancer, Gastrectomy, Laparoscopy, PET-CT, Radiopharmaceuticals

Background

Gastric cancer is the third leading cause of cancer-related death worldwide [1]. In most Western countries, curative treatment for locally advanced gastric cancer includes perioperative chemotherapy and gastrectomy [2–5]. Despite this multimodal approach, the prognosis remains poor, with five-year survival rates ranging from 36–45% [3–5]. Key reasons for curative treatment failure include tumor irresectability and/or unexpected peritoneal metastases identified at the onset of surgery, and the appearance of locoregional recurrences or distant metastases during neoadjuvant chemotherapy or shortly after surgery [3, 4]. In the Netherlands, 40–60% of patients undergoing surgery with curative intent develop disease recurrence [6]. When peritoneal and/or distant metastases or irresectable disease are identified, the treatment intent shifts from curative to palliative, and gastrectomy is not deemed useful anymore. This highlights the importance of accurate staging to avoid unnecessary toxic and costly systemic treatments and/or surgery.

The current diagnostic workup consists of a gastroscopy, a computed tomography (CT) scan, and a staging laparoscopy (SL). However, CT scans often fail to detect metastases, especially peritoneal metastases, and while SL adds value in identifying peritoneal tumor deposits and locally irresectable disease, it is an invasive procedure with associated complication risks [7]. Moreover, SL cannot detect metastases outside the abdominal cavity. Positron Emission Tomography/CT with 2-[^{18}F]fluoro-2-deoxy-D-glucose ([^{18}F]FDG PET/CT) was shown to have very limited added value in detecting distant metastases, particularly in gastric cancer types with low glucose uptake [7]. Therefore, there remains a need for a reliable, non-invasive diagnostic tool to detect distant and peritoneal metastases in patients with locally advanced gastric cancer.

New radiotracers targeting tumor stroma have been developed to overcome the challenges posed by low [^{18}F] FDG-avid cancer types. Highly promising tracers focus on fibroblast activation protein (FAP), which is expressed in the tumor stroma of over 90% of epithelial cancers and is nearly absent in healthy tissues [8]. FAP-specific small molecule inhibitors (FAPI) have been developed for PET/CT imaging, and preliminary Asian studies using ^{68}Gallium(^{68}Ga)-labeled FAPI PET/CT in gastric cancer have demonstrated improved diagnostic sensitivity, tracer uptake and tumor-to-background ratio compared to [^{18}F]FDG PET/CT [9–15]. However, these results may not be directly applicable to Western patients due to different biological characteristics between Asian and non-Asian gastric cancers [16–18]. Additionally, these studies used ^{68}Ga-labeled FAPI-04, whereas ^{18}F-labeled FAPI-74 may offer several advantages. These include improved sensitivity in detecting smaller lesions due to better spatial resolution and image quality, resulting from its shorter mean positron range in soft tissue (~0.6 mm) compared to ^{68}Ga FAPI-04 (~3 mm). Moreover, while an expensive locally installed generator is needed for production of ^{68}Ga FAPI-04, ^{18}F FAPI-74 can be centrally produced, since its longer half-life time compared to ^{68}Ga FAPI-04 (110 versus 68 min) enables widespread distribution and use. Additionally, ^{18}F FAPI-74 emits lower energy positrons (97%, maximum ~ 634 keV) compared to ^{68}Ga FAPI-04 (88%, maximum energy ~ 1899 keV), reducing radiation exposure to patients. Furthermore, the Asian studies involved small sample sizes, underscoring the need for larger, more homogeneous trials in Western patient populations using [^{18}F]AlF-FAPI-74 to validate these preliminary findings and translate them to the European population.

Methods

Aim of the study

This study aims to assess the applicability and impact of [^{18}F]AlF-FAPI-74 as an imaging biomarker for accurately detecting both distant and peritoneal metastases in patients with locally advanced gastric cancer (cT3-4 and/or cN+) after staging by gastroscopy and CT in a Western gastric cancer cohort.

Study design

The PLASTIC-3 study is a Dutch, multicenter, prospective, clinical trial. All patients with a locally advanced gastric adenocarcinoma who are candidates for a gastrectomy with curative intent will be invited to participate in this study. A locally advanced gastric adenocarcinoma is defined as a transmural tumor invading the outer layer of the stomach (cT3-4 according to the 8th edition of the American Joint Committee on Cancer Tumor, Node, Metastases (TNM) staging system), with or without regional lymph node metastasis in 1 or more lymph nodes (cN+) as objectified on CT [19]. The hypothesis of this study is that adding [^{18}F]AlF-FAPI-74 PET/CT to the staging algorithm will identify at least two-thirds of all patients that are metastasized or found incurable during SL, neoadjuvant treatment or at gastrectomy. This could improve the detection of incurable disease and prevent patients from undergoing futile and costly SL, gastrectomies and/or toxic systemic treatments. This may reduce patient burden, enhance quality of life, and potentially lower healthcare costs.

Table 1 Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria
Histologically proven adenocarcinoma of the stomach or the esophagogastric junction (Siewert type III), by gastroscopy	Siewert type I-II esophagogastric junction tumor
Age ≥ 18 years	Unfit or unwilling to undergo study procedures
Surgically resectable, advanced tumor (cT1-2N1-3M0 or cT3-4bN0-3M0), as determined on gastroscopy and contrast-enhanced CT of thorax and abdomen. Intention to perform a gastrectomy, based on a multidisciplinary team meeting and shared decision making	Unfit or unwilling to undergo surgery
Patients must have given written informed consent	Pregnancy at time of the [^{18}F]AlF-FAPI-74 PET/CT scan
Patients who have recently participated in an interventional study with an investigational medicinal product may only participate if an interaction with study procedures is deemed unlikely by the study team (e.g. based on mechanism or washout period)	Medical or mental conditions that compromise the patient's ability to give informed consent
	Illiterate patients unable to complete the resource use and quality of life questionnaires
	Inability to undergo PET/CT scans due to factors such as claustrophobia, weight limits, or the inability to lie flat for the duration of the scan (approximately 30 min)

Study population

The study population consists of patients with locally advanced gastric adenocarcinoma (cT1-2N1-3M0 or cT3-4bN0-3M0) suitable for surgery, after initial staging with gastroscopy and contrast-enhanced CT of the thorax and abdomen. Patients' inclusion and exclusion criteria are defined in Table 1.

Study protocol

Initial staging

Initial staging should be performed according to national guidelines, including a gastroscopy with tumor biopsies and a contrast-enhanced diagnostic CT scan of the thorax and abdomen [20]. After initial staging, patients will be discussed in the multidisciplinary team (MDT). In case of a cT3-4 tumor and/or N+disease, which is deemed to be eligible for resection with curative intent, patients may be approached to participate in this study (Table 2) [19].

Table 2 Clinical TNM-staging according to the 8th edition of the AJCC gastric cancer staging system [19]

T – Primary tumor	
Tx	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	Carcinoma in situ: intraepithelial tumor without invasion of the lamina propria
T1a	Tumor invades lamina propria, muscularis mucosae or submucosa
T1b	Tumor invades submucosa
T2	Tumor invades muscularis propria
T3	Tumor penetrates subserosal connective tissue without invasion of visceral peritoneum or adjacent structures. T3 tumors also include those extending into the gastrocolic or gastrohepatic ligaments, or into the greater or lesser omentum, without perforation of the visceral peritoneum covering these structures
T4a	Tumor invades serosa (visceral peritoneum)
T4b	Tumor invades adjacent structures (spleen, transverse colon, liver, diaphragm, pancreas, abdominal wall, adrenal gland, kidney, small intestine, and retroperitoneum)
N – Regional lymph nodes	
Nx	Regional lymph node(s) cannot be assessed
N0	No regional lymph node metastasis
N1	Metastasis in 1 to 2 regional lymph nodes
N2	Metastasis in 3 to 6 regional lymph nodes
N3	Metastasis in 7 or more regional lymph nodes
N3a	Metastasis in 7–15 regional lymph nodes
N3b	Metastasis in 16 or more regional lymph nodes
M – Distant metastasis	
M0	No distant metastasis
M1	Distant metastasis

Differentiation between cT2 and cT3 tumors is not always possible with initial staging. In case there is considerable doubt whether a tumor is cT2 or cT3, patients may be included if deemed appropriate by the MDT. Treatment of patients will be performed according to standard of care as recommended by the Dutch national guideline [20].

Patient inclusion

Patients will be informed and enrolled at the surgical outpatient department of one of the Dutch investigational centers. Since participation in this study requires undergoing additional study interventions beyond standard care, this study falls under the Dutch Medical Research Involving Human Subjects Act (WMO). Patients must provide informed consent, confirming their awareness that data will be used anonymously for research purposes. They will also agree to complete quality of life, resource use, and patient reported experience measure (PREM) questionnaires via the existing infrastructure of the "Prospective Observational Cohort Study of Oesophageal-gastric cancer Patients" (POCOP) or the coordinating investigator [21].

[^{18}F]AlF-FAPI-74 PET/CT

Patients are scanned ~ 60 min (range 55–70 min) after intravenous injection of approximately 3.0 MBq/kg of [^{18}F]AlF-FAPI-74 in each participating center on an EARL accredited PET/CT scanner, ensuring comparability and accurate quantification of tracer uptake [22–24]. First a low-dose unenhanced CT from vertex to the proximal femora is performed. To minimize bladder filling during [^{18}F]AlF-FAPI-74 scanning, subsequent PET-acquisition starts from mid-femoral toward the head. The [^{18}F]AlF-FAPI-74 PET/CT procedures and assessments, and other imaging measurements will be performed in a standardized manner, according to the Dutch acquisition of FAPI PET/CT protocol [25]. After tracer injection and during the scan, patients are observed for side effects (no side effects have been reported to date).

Scans are read, interpreted, and reported by or under supervision of an experienced nuclear medicine physician, supported by any additional imaging performed during routine clinical workup.

If the [^{18}F]AlF-FAPI-74 PET/CT scan is positive for distant and/or peritoneal metastases, patients undergo biopsy/cytology or additional imaging to confirm those suspect lesions. If the [^{18}F]AlF-FAPI-74 PET/CT scan is negative/unconfirmed for distant and/or peritoneal metastases, patients will undergo SL with biopsy, according to standard care (Figs. 1 and 2). When additional tissue examination and/or additional imaging after a positive [^{18}F]AlF-FAPI-74 PET/CT scan confirms metastases, patients do not need to undergo an additional SL and the treatment plan changes from curative to palliative. The same applies for patients found to have peritoneal metastases at SL after a negative or unconfirmed [^{18}F]AlF-FAPI-74 PET/CT scan.

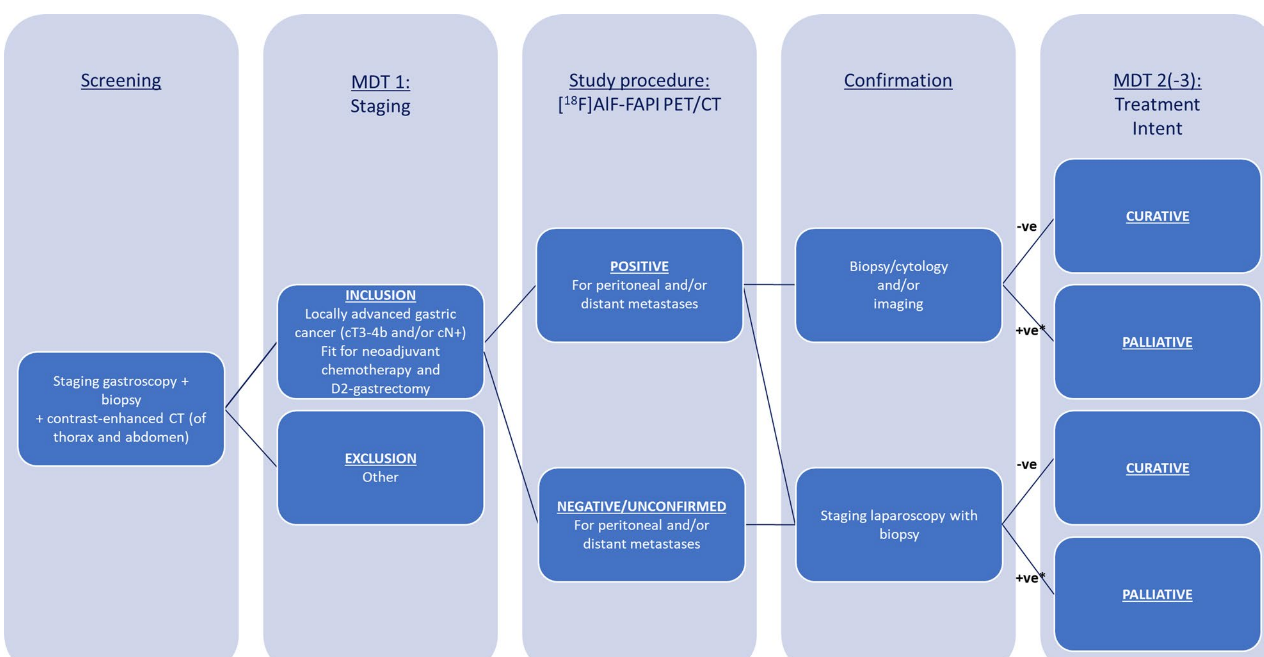

Fig. 1 Flowchart of the PLASTIC-3 study. An investigational [^{18}F]AlF-FAPI-74 PET/CT will be performed before staging laparoscopy. In case of positive findings during [^{18}F]AlF-FAPI-74 PET/CT, additional biopsy/cytology or imaging will be performed, and futile staging laparoscopy with biopsy will be omitted. MDT: multidisciplinary team. *In case of oligometastatic disease, the MDT might decide to treat the patient with curative intent

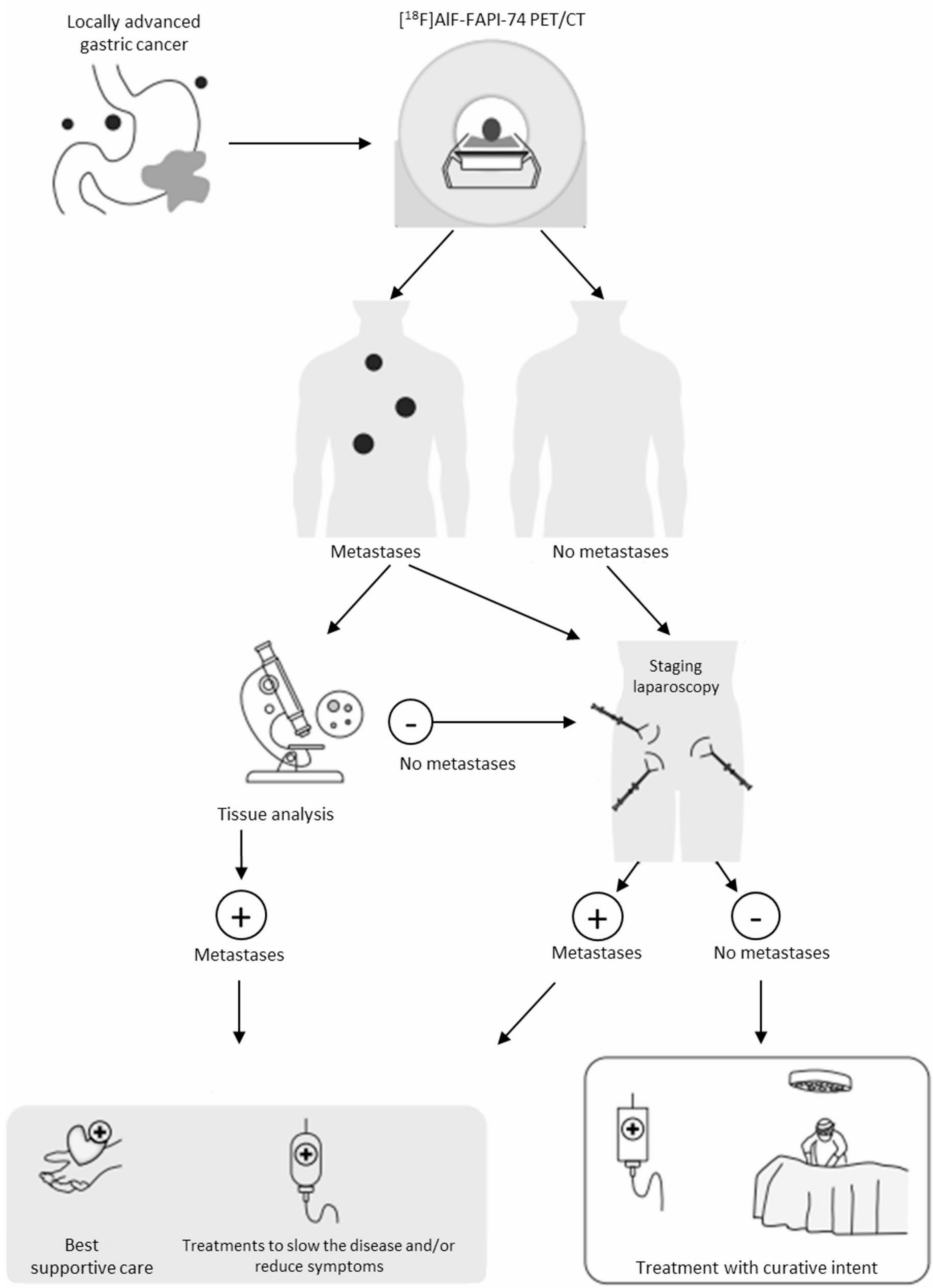

Fig. 2 Study design of the PLASTIC-3 study: different phases of the study

Staging laparoscopy

SL will be performed after negative or unconfirmed lesions at [^{18}F]AlF-FAPI-74 PET/CT, before the initiation of treatment, and should be executed or supervised by an experienced gastrointestinal or oncological surgeon. During SL, the resectability of the primary tumor (cT-stage) and the presence or absence of peritoneal metastases need to be evaluated. To evaluate resectability of the tumor, a thorough inspection of the stomach and tumor should be performed according to national standards. Based on a recent Dutch nationwide Delphi consensus study regarding the indications for and execution of SL in patients with gastric cancer, this includes inspection of infiltration of the tumor in surrounding structures or organs (diaphragm, esophagus, spleen, pancreas, and posterior vascular structures, mesentery of the colon) only when preoperative radiological imaging suggests ingrowth [26]. Regardless of prior suspicion of ingrowth, the left lobe of the liver should always be inspected since no additional dissection needs to be performed to judge this. To evaluate the presence or absence of peritoneal metastases, all quadrants of the peritoneal cavity should be inspected. The abdomen should systematically be inspected according to the regions of the peritoneal cancer index (PCI) of Sugarbaker et al. and the complete peritoneal cancer index score should be documented as mentioned in the standardized operation report (Supplementary File 1) [27]. Only in patients in whom there is a preoperative suspicion of tumor on imaging, it is advised to open the omental bursa and inspect it accordingly. In case of suspicious macroscopic lesions, biopsies will be taken and sent for pathological review. In addition, standard peritoneal lavage and/or aspiration of ascites during SL for cytology needs to be performed. Peritoneal lavage consists of a minimum total volume of 200cc saline which should be introduced and equally dispersed throughout the peritoneal cavity, the left and right subphrenic space and Douglas cavity, according to the AJCC protocol [19, 26]. After collection, the samples will be sent for pathological review.

Multidisciplinary team (MDT) meeting

Ideally, patients will be discussed in a first MDT meeting after the initial staging with gastroscopy and CT of the thorax and abdomen, and again in a second MDT meeting following additional staging with [^{18}F]AlF-FAPI-74 PET/CT and SL (Fig. 1). In practice, some patients may skip either the first or second MDT meeting and proceed directly to additional staging or treatment without further MDT involvement. However, each included patient should be discussed in at least one MDT meeting. Occasionally, an additional diagnostic modality may be needed, requiring the patient to be discussed in a third MDT meeting.

Treatment

After completion of full staging, and if the tumor is deemed to be resectable, patients will be scheduled for treatment. No treatment with neoadjuvant chemotherapy or surgery should be performed before completion of staging. Treatment of patients will be performed according to standard of care as recommended by the Dutch national guideline, with no restrictions regarding any specific treatment strategy, such as the type of (neo)adjuvant treatment regimen (e.g., chemotherapy, targeted therapy, immunotherapy), or type/approach of resection) [20]. Although positive peritoneal cytology without macroscopic peritoneal disease is considered M1-disease according to the TNM-8, no treatment restrictions of these patients will be imposed in the study protocol due to limited evidence [28].

Outcome measures

The primary outcome of this study is the proportion of patients in whom the [^{18}F]AlF-FAPI-74 PET/CT scan leads to a change in treatment intent as determined by the local multidisciplinary team (MDT) meeting. This includes the number of prevented unnecessary surgeries (SL and/or gastrectomies) and the number of changes from curative to palliative treatment. An additional primary outcome measure is the proportion of patients in whom [^{18}F]AlF-FAPI-74 PET/CT leads to changes in the diagnostic workup determined by the local MDT meeting, that will take confirmative SL, biopsies and (follow-up) imaging into account. This outcome measure also includes the number of additional biopsies or longitudinal imaging, and the number of changes in the extent of surgery.

Secondary outcomes include the diagnostic performance, measured in sensitivity, specificity, diagnostic accuracy, positive predictive value, and negative predictive value. The composite reference standard consists of histopathologic tumor tissue collected during biopsy and SL, (follow-up) imaging, intra-operative findings during curative gastrectomy and 1-year follow-up after completion of full staging. In addition, outcomes include the occurrence and impact of incidental and/or non-specific findings on [^{18}F]AlF-FAPI-74 PET/CT, diagnostic time delay, safety regarding the clinical use of [^{18}F]AlF-FAPI-74 PET/CT, quality of life, additional patient burden, and its cost-effectiveness.

Sample size calculation

The primary objective concerns the question whether clinical implementation of [^{18}F]AlF-FAPI-74 PET/CT can contribute significantly to staging of locally advanced gastric cancer. We hypothesize that, to be of clinical relevance, full implementation of [^{18}F]AlF-FAPI-74 PET/CT should be able to detect at least two-thirds (67%) of all patients that are found to have metastatic disease or that are otherwise found incurable during SL, neoadjuvant chemotherapy or at gastrectomy. In the PLASTIC study, occult metastases led to a change in treatment intent in 16.5% (65/394) of subjects [7, 29]. Expecting that [^{18}F]AlF-FAPI-74 PET/CT detects ≥85% in our cohort, we would need to have complete data of 37 subjects with occult metastases to be able to reject our null-hypothesis that we detect less than 67% of occult metastasis (one sample binomial test, one-sided test alpha of < 5%, power > 80%, G*Power, version 3.1.9.6). This would imply 37/0.165 = 225 subjects with complete data in total. Assuming incomplete data in 10% (withdrawn consent, loss to follow-up, missing data), we would need to include 250 patients. Based on patient numbers in participating centers and previous experience with PLASTIC (394 patient inclusions in 2.5 years), we expect inclusion of 250 patients to be feasible in three years [7].

Statistical analysis

The primary outcome measures, proportion of patients staged as cM1-disease based on [^{18}F]AlF-FAPI-74 PET/CT, proportion of prevented unnecessary surgeries, and proportion of patients in whom [^{18}F]AlF-FAPI-74 PET/CT leads to changes in the diagnostic workup determined by the local MDT meetings, will be presented as percentages. Additionally, the proportion of additional biopsies or longitudinal imaging, and the proportion of changes in extent of surgery will be presented as percentages.

The diagnostic performance of [^{18}F]AlF-FAPI-74 PET/CT, measured in accuracy, sensitivity, specificity, positive and negative predictive value, to detect M1-disease will be evaluated using contingency tables. The reference for a positive [^{18}F]AlF-FAPI-74 PET/CT will be confirmation by tissue sampling, or by imaging if sampling is not feasible or unsafe. The reference for a negative [^{18}F]AlF-FAPI-74 PET/CT will be SL for peritoneal lesions, supplemented with serial imaging for distant metastases and intraoperative findings during gastrectomy.

All proportions will be presented including their 95% Clopper-Pearson interval as (binomial) confidence limits.

Receiver operator characteristic (ROC) curve analysis will be used to describe the diagnostic performance of non-binary measurements including the mean standardized uptake value corrected for lean body mass (SUL_{mean}), PCI and volumetric parameters, and an optimal cut-off will be described which best selects patients not benefitting from SL. The potential diagnostic time delay occurring due to the extra investigational [^{18}F]AlF-FAPI-74 PET/CT will be calculated as the time in days between pre-diagnostic and post-diagnostic MDT meetings.

Cost-effectiveness and quality of life (QoL) analysis

The cost-effectiveness analysis will only be conducted in the event that the [^{18}F]AlF-FAPI-74 PET/CT is found to have a clinical benefit (via the primary outcome) throughout the course of the study, and will be assessed by comparing the data from the PLASTIC-3 study with the cost-effectiveness data from the original PLASTIC study [7]. Propensity score matching or similar statistical analysis will be used to address potential bias of using two independent prospective observational cohort studies [30]. The extrapolation of trial-based data will be performed in R using the bootstrap percentile method combined with (multiple) imputation using predictive mean matching. Results will be visualized in incremental cost-effectiveness planes and cost-effectiveness acceptability curves. Aligned with the PLASTIC study, generic quality of life instrument EQ-5D-5L and the disease specific EORTC QLQ-C30 questionnaire, and resource use cost questionnaires will be prospectively sent to participants before staging as baseline, after completion of full staging but before starting the treatment, and after completion of full staging at 3, 6, 9 and 12 months [31–34]. Health-related quality of life outcomes will be reported as utilities using the Dutch population value set [35]. Hence, quality-adjusted life years (QALY's) can be calculated and compared between the two groups. A PREM-questionnaire was developed in collaboration with patient representatives to measure the burden of [^{18}F]AlF-FAPI-74 PET/CT and SL, and will be presented to patients after completing the full diagnostic workup.

Time schedule

The study recently started on July 3rd 2025. The first 36 months will consist of inclusion of patients, followed by a 12-month patient follow-up period after completion of staging. The study is expected to conclude in July 2029, 48 months after initiation.

Discussion

The PLASTIC-3 study is a Dutch, multicenter, prospective, clinical trial designed to evaluate the value and impact of [^{18}F]AlF-FAPI-74 PET/CT for detecting cM1-disease in addition to the staging process with gastroscopy, CT, and SL, in patients with locally advanced gastric cancer.

The Dutch multicenter PLASTIC study demonstrated that [^{18}F]FDG PET/CT has very limited value in detecting distant metastases in locally advanced gastric cancer [7]. Quantitative multiparametric radiomics analysis did not enhance preoperative detection of distant and peritoneal metastases [36]. The sensitivity and specificity of [^{18}F]FDG PET/CT in detecting gastric cancer are impacted by factors such as physiological stomach uptake, gastritis-related uptake, and varying [^{18}F]FDG avidity across histologic subtypes. Notably, diffuse-type, mucinous, and signet ring cell tumors exhibit low [^{18}F]FDG uptake, likely due to low tumor cellularity and reduced glucose-transporter-1 expression [37]. Additionally, [^{18}F]FDG PET/CT has lower sensitivity for detecting peritoneal metastases, likely due to the typically small size of these lesions, physiological gut uptake, and PET/CT misalignment from gut movement [38]. [^{18}F]FDG PET/CT was therefore recently de-implemented from the Dutch guidelines. In contrast, PET/CT using different FAP-radiopharmaceuticals for detecting distant and peritoneal metastases in patients with locally advanced gastric cancer show promising results [9–15]. However, none of these previous studies used the [^{18}F]AlF-FAPI-74 tracer, which may offer improved detection of small peritoneal lesions compared to [^{68}Ga]Ga-FAPI.

In this study design, [^{18}F]AlF-FAPI-74 PET/CT readers are blinded for the SL outcome, but not vice versa. This approach allows findings on [^{18}F]AlF-FAPI-74 PET/CT to be confirmed during SL and enables surgeons to specifically look for and examine lesions detected on [^{18}F]AlF-FAPI-74 PET/CT, thereby reducing the risk of false positives. SL will only be omitted in cases where [^{18}F]AlF-FAPI-74 PET/CT indicates metastatic disease that is subsequently confirmed by an alternative procedure, such as an image-guided biopsy, since it is considered unethical to perform an unnecessary SL in the presence of proven metastases. However, peritoneal metastases will not be confirmed in such cases; therefore, final decisions will be made by the local MDT meeting. To account for interobserver variability, blinded double-readings of the abdominal part of the scan will be performed in one center. Additionally, the learning curve of [^{18}F]AlF-FAPI-74 PET/CT assessment will be evaluated by comparing sensitivity and specificity between the first and last 75 included patients.

One potential downside of [^{18}F]AlF-FAPI-74 PET/CT is the detection of incidental or non-specific findings, which may necessitate additional examinations, increasing costs and potential diagnostic time delay. Therefore, this study will also assess the patient burden and the cost-effectiveness of [^{18}F]AlF-FAPI-74 PET/CT compared to SL.

The current study is relevant because, if [^{18}F]AlF-FAPI-74 PET/CT demonstrates clinical utility, its implementation into clinical practice could enable a more accurate, patient-centered, and cost-effective non-invasive diagnostic approach. Improved diagnostic accuracy may reduce unnecessary exposure to invasive and toxic treatments, allowing more patients to benefit from optimal treatment strategies. The study's findings are expected to invoke updates to the international guidelines on gastric cancer treatment, in countries with similar treatment and healthcare systems. Future clinical trials could explore theranostic treatment strategies for distant or peritoneal metastases using FAPI tracers, although further research on these potential applications is warranted.

Abbreviations

CT	Computed tomography
EARL	EANM Research Ltd.
FAP	Fibroblast activation protein
FAPI	Fibroblast activation protein inhibitor
FDG	Fluorodeoxyglucose
FDG PET	Fluorodeoxyglucose positron emission tomography
KWF	Dutch Cancer Society – Koningin Wilhelmina Fonds voor kankerbestrijding
MDT	Multidisciplinary team
METC	Medical ethical review board
M1-disease	Peritoneal and/or extra-abdominal distant metastases
PCI	Peritoneal cancer index
PET	Positron emission tomography
PLASTIC-3	[^{18}F]F-FAPI PET/CT and Laparoscopy in STagIng advanced gastric Cancer: a multicenter prospective study
PREM	Patient reported experience measure
QoL	Quality of life
ROC	Receiving operating characteristic
SL	Staging laparoscopy
SUL	Standardized uptake value corrected for lean body mass
TNM	Tumor, Node, Metastases
WMO	Medical research involving human subjects act

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12885-025-15347-7.

Supplementary Material 1. Standardized operation report for staging laparoscopy for gastric cancer.

Acknowledgements

We would like to thank Liesbeth Timmermans and Andre Bijeman (Patient Federation SPKS) and Kim Holtzer (Dutch Federation of Cancer Patient Organisations) for their contribution to the design of this study.

Authors' contributions

This manuscript is published on behalf of all members of the PLASTIC-3 Study Group. All authors were involved in developing the original study and protocols. EV, WEM, LJACH, SS, MS, VK, ASLPC, JJG, MP, PDS, RJB, MIBH, MJD, BJE, HHH, MGEHL, NMLS, MDPL, WN, ECO, WJGO, JWS, WS, WOS, JHMBS, LEW, BPLW, BPLW are responsible for the clinical input. LT, SWJMS, DV, RvH, JPR, LFG drafted the manuscript. All authors provided significant input to the manuscript by means of revisions and have read and approved the final manuscript.

Funding

This trial is supported by a Themacall Biomarkers Grant (15400) of the Dutch Cancer Society – Koningin Wilhelmina Fonds (KWF) voor kankerbestrijding. There is no role for the funding party in collecting, analyzing and interpreting data, and in writing the manuscript. These tasks will be fulfilled by the study coordinator, on behalf of all members of the PğLASTIC-3 Study Group.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study was approved by the Medical Research Ethics Committee (MREC, Dutch: METC) Leiden-The Hague-Delft (reference number: P25.012) on May 5th 2025. As this study does allocate patients to study interventions other than usual care, as recommended by the Dutch guidelines, this study does fall within the Medical Research Involving Human Subjects Act (WMO). Written informed consent will be obtained from all study participants.

Consent for publication
Not applicable.

Competing interests
MIBH is consultant for Intuitive, Johnson and Johnson, Medtronic, Stryker and BBraun (all fees paid to institution). LJACH received consultant fees and is member scientific advisory board of InnoSer Biosciences (not related). MGEHL is consultant and research support for Terumo, Boston Scientific, consultant for OncoSil, Becton and Dickinson and Novartis. MDPL received grants from Medtronic and Galvani for research. All other authors declare they have no competing interests.

Author details
[1]Department of Surgery, University Medical Center Utrecht, Utrecht, The Netherlands
[2]Department of Radiology, Leiden University Medical Center, Leiden, The Netherlands
[3]Department of Medical Imaging, Radboud University Medical Center, Nijmegen, The Netherlands
[4]Department of Radiology and Nuclear Medicine, Erasmus MC University Medical Center Rotterdam, Rotterdam, The Netherlands
[5]Department of Surgery, Leiden University Medical Center, Leiden, The Netherlands
[6]Department of Gastroenterology and Hepatology, Leiden University Medical Center, Leiden, The Netherlands
[7]Department of Health Technology and Services Research, Faculty of Behavioural Management and Social Sciences, University of Twente, Enschede, The Netherlands
[8]Department of Medical Oncology, Leiden University Medical Center, Leiden, The Netherlands
[9]Department of Cell and Chemical Biology, Leiden University Medical Center, Leiden, The Netherlands
[10]Department of Pathology, Leiden University Medical Center, Leiden, The Netherlands
[11]Department of Biomedical Data Sciences, Leiden University Medical Center, Leiden, The Netherlands
[12]Department of Clinical Pharmacy and Toxicology, Leiden University Medical Center, Leiden, The Netherlands
[13]Department of Surgery, Amsterdam UMC Location University of Amsterdam, Amsterdam, The Netherlands
[14]Cancer Center Amsterdam, Cancer Treatment and Quality of Life, Amsterdam, The Netherlands
[15]Department of Surgery, Ziekenhuisgroep Twente, Almelo, The Netherlands
[16]Department of Surgery, University Medical Center Groningen, Groningen, The Netherlands
[17]Department of Radiology and Nuclear Medicine, UMC Utrecht, Utrecht, The Netherlands
[18]Department of Nuclear Medicine, Catharina Hospital Eindhoven, Eindhoven, The Netherlands
[19]Department of Surgery, Catharina Hospital Eindhoven, Eindhoven, The Netherlands
[20]Department Nuclear Medicine and Molecular Imaging, University Medical Center Groningen, Groningen, The Netherlands
[21]Department of Nuclear Medicine, Netherlands Cancer Institute – Antoni van Leeuwenhoek Hospital, Amsterdam, The Netherlands
[22]Department of Radiology and Nuclear Medicine, Rijnstate Hospital, Arnhem, The Netherlands
[23]Department of Biomedical Sciences and Humanitas Clinical and Research Center, Humanitas University, Milan, Italy
[24]Department of Surgical Oncology, Netherlands Cancer Institute – Antoni van Leeuwenhoek Hospital, Amsterdam, The Netherlands
[25]Department of Nuclear Medicine, Zuyderland Medical Center, Heerlen, The Netherlands
[26]Department of Surgery, Zuyderland Medical Center, Heerlen, The Netherlands
[27]Department of Nuclear Medicine, Ziekenhuisgroep Twente, Almelo, The Netherlands
[28]Department of Surgery, Erasmus MC Cancer Institute, Erasmus University Medical Center, Rotterdam, The Netherlands
[29]Department of Surgery, Rijnstate Hospital, Arnhem, The Netherlands
[30]Biomedical Photonic Imaging Group, University of Twente, Drienerlolaan 5, Enschede, NB 7522, The Netherlands

Received: 9 October 2025 / Accepted: 13 November 2025
Published online: 27 November 2025

References
1. Sung H, Ferlay J, Siegel RL, Laversanne M, Soerjomataram I, Jemal A, et al. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin. 2021;71(3):209–49.
2. Songun I, Putter H, Kranenbarg EMK, Sasako M, van de Velde CJH. Surgical treatment of gastric cancer: 15-year follow-up results of the randomised nationwide Dutch D1D2 trial. Lancet Oncol. 2010;11(5):439–49.
3. Al-Batran SE, Homann N, Pauligk C, Goetze TO, Meiler J, Kasper S, et al. Perioperative chemotherapy with fluorouracil plus leucovorin, oxaliplatin, and docetaxel versus fluorouracil or capecitabine plus cisplatin and epirubicin for locally advanced, resectable gastric or gastro-oesophageal junction adenocarcinoma (FLOT4): a randomised, phase 2/3 trial. Lancet. 2019;393(10184):1948–57.
4. Cunningham D, Allum WH, Stenning SP. Perioperative chemotherapy versus surgery alone for resectable gastroesophageal cancer. N Engl J Med. 2006;355:11–20.
5. Cunningham D, Starling N, Rao S, Iveson T, Nicolson M, Coxon F, et al. Capecitabine and oxaliplatin for advanced esophagogastric cancer. N Engl J Med. 2008;358(1):36–46.
6. Oncoline. Landelijke werkgroep gastro-intestinale tumoren versie: 2.2. 2017. 2017. Available from: https://www.oncoline.nl/maagcarcinoom. Cited 2020 Aug 31.
7. Gertsen EC, Brenkman HJF, Van Hillegersberg R, Van Sandick JW, Van Berge Henegouwen MI, Gisbertz SS, et al. 18F-fludeoxyglucose-positron emission tomography/computed tomography and laparoscopy for staging of locally advanced gastric cancer: a multicenter prospective Dutch cohort study (PLASTIC). JAMA Surg. 2021;156(12):E1-9.
8. Lindner T, Loktev A, Altmann A, Giesel F, Kratochwil C, Debus J, et al. Development of quinoline-based theranostic ligands for the targeting of fibroblast activation protein. J Nucl Med. 2018;59(9):1415–22.
9. Chen H, Pang Y, Wu J, Zhao L, Hao B, Wu J, et al. Comparison of [68Ga]Ga-DOTA-FAPI-04 and [18F] FDG PET/CT for the diagnosis of primary and metastatic lesions in patients with various types of cancer. Eur J Nucl Med Mol Imaging. 2020;47(8):1820–32.
10. Gündoğan C, Kömek H, Can C, Yildirim ÖA, Kaplan İ, Erdur E, et al. Comparison of 18F-FDG PET/CT and 68Ga-FAPI-04 PET/CT in the staging and restaging of gastric adenocarcinoma. Nucl Med Commun. 2022;43(1):64–72.
11. Jiang D, Chen X, You Z, Wang H, Zhang X, Li X, et al. Comparison of [68 Ga]Ga-FAPI-04 and [18F]-FDG for the detection of primary and metastatic lesions in patients with gastric cancer: a bicentric retrospective study. Eur J Nucl Med Mol Imaging. 2022;49(2):732–42.
12. Lin R, Lin Z, Chen Z, Zheng S, Zhang J, Zang J, et al. [68Ga]Ga-DOTA-FAPI-04 PET/CT in the evaluation of gastric cancer: comparison with [18F]FDG PET/CT. Eur J Nucl Med Mol Imaging. 2022;49(8):2960–71.
13. Kuten J, Levine C, Shamni O, Pelles S, Wolf I, Lahat G, et al. Head-to-head comparison of [68Ga]Ga-FAPI-04 and [18F]-FDG PET/CT in evaluating the extent of disease in gastric adenocarcinoma. Eur J Nucl Med Mol Imaging. 2022;49(2):743–50.
14. Zhang S, Wang W, Xu T, Ding H, Li Y, Liu H, et al. Comparison of diagnostic efficacy of [68Ga]Ga-FAPI-04 and [18F]FDG PET/CT for staging and restaging of gastric cancer. Front Oncol. 2022;1:12.
15. Miao Y, Feng R, Guo R, Huang X, Hai W, Li J, et al. Utility of [68Ga]FAPI-04 and [18 F]FDG dual-tracer PET/CT in the initial evaluation of gastric cancer. Eur Radiol. 2023;33:4355–66.
16. Russo A, Li P, Strong VE. Differences in the multimodal treatment of gastric cancer: east versus west. J Surg Oncol. 2017;115:603–14.
17. Chan WL, Lam KO, Lee VHF, Davidson M, So TH, Li JS, et al. Gastric cancer – from aetiology to management: differences between the east and the west. Clin Oncol. 2019;31(8):570–7.

18. Lin SJ, Gagnon-Bartsch JA, Tan IB, Earle S, Ruff L, Pettinger K, et al. Signatures of tumour immunity distinguish Asian and non-Asian gastric adenocarcinomas. Gut. 2015;64(11):1721–31.
19. American Joint Committee on Cancer. AJCC cancer staging manual. 8th ed. 2017.
20. Federatie Medisch Specialisten. Richtlijn maagcarcinoom. 2023.
21. Coebergh van den Braak R, van Rijssen L, van Kleef J, Vink G, Berbee M, van Berge Henegouwen M, et al. Nationwide comprehensive gastro-intestinal cancer cohorts: the 3P initiative. Acta Oncol. 2017;57(2):195–202.
22. Rahmim A, Lodge MA, Karakatsanis NA, Panin VY, Zhou Y, McMillan A, et al. Dynamic whole-body PET imaging: principles, potentials and applications. Eur J Nucl Med Mol Imaging. 2019;46:501–18 (Springer Berlin Heidelberg).
23. Giesel FL, Adeberg S, Syed M, Lindner T, Jiménez-Franco LD, Mavriopoulou E, et al. FAPI-74 PET/CT using either 18F-AlF or cold-kit 68Ga labeling: biodistribution, radiation dosimetry, and tumor delineation in lung cancer patients. J Nucl Med. 2021;62(2):201–7.
24. Watabe T, Naka S, Tatsumi M, Kamiya T, Kimura T, Shintani Y, et al. Initial evaluation of [18F]FAPI-74 PET for various histopathologically confirmed cancers and benign lesions. J Nucl Med. 2023;64(8):1225–31.
25. Slart RHJA, Kool W, Zwezerijnen GJC, van den Hoven AF. Acquisitie protocol NVNG FAPI werkgroep [Internet]. NVNG; 2023 Jun 21 [cited 2025 Sept 26]. Available from: https://www.fapiwerkgroep.nl/docs/FAPI%20PET-CT%20acquisitie%20protocol_NVNG%20FAPI%20werkgroep_v001_21-06-2023.pdf
26. van der Sluis K, Guchelaar NAD, Triemstra L, Mathijssen RHJ, Ruurda JP, Wijnhoven BPL, et al. Staging laparoscopy in gastric cancer patients: From a Dutch nationwide Delphi consensus towards a standardized protocol. Eur J Surg Oncol. 2024 Jun;50(6):108278. https://doi.org/10.1016/j.ejso.2024.108278.
27. Jacquet P, Sugarbaker PH. Clinical research methodologies in diagnosis and staging of patients with peritoneal carcinomatosis. Cancer Treat Res. 1996;82:359–74.
28. Jamel S, Markar SR, Malietzis G, Acharya A, Athanasiou T, Hanna GB. Prognostic significance of peritoneal lavage cytology in staging gastric cancer: systematic review and meta-analysis. Gastric Cancer. 2018;21(1):10–8.
29. Smyth E, Schöder H, Strong VE, Capanu M, Kelsen DP, Coit DG, et al. A prospective evaluation of the utility of 2-deoxy-2-[18F]fluoro-D-glucose positron emission tomography and computed tomography in staging locally advanced gastric cancer. Cancer. 2012;118(22):5481–8.
30. Austin PC. An introduction to propensity score methods for reducing the effects of confounding in observational studies. Multivar Behav Res. 2011;46(3):399–424.
31. Aaronson N, Ahmedzai S, Bergman B, Bullinger M, Cull A, Duez N, et al. The European organization for research and treatment of cancer QLQ-C30: a quality-of-life instrument for use in international clinical trials in oncology. J Natl Cancer Inst. 1993;85(5):365–76.
32. Herdman M, Gudex C, Lloyd A, Janssen M, Kind P, Parkin D, et al. Development and preliminary testing of the new five-level version of EQ-5D (EQ-5D-5L). Qual Life Res. 2011;20(10):1727–36.
33. iMTA Productivity and Health Research Group. Manual iMTA Productivity Cost Questionnaire (iPCQ). Rotterdam: iMTA, Erasmus University Rotterdam; 2018.
34. iMTA Productivity and Health Research Group. Manual iMTA Medical Cost Questionnaire (iMCQ). Rotterdam: iMTA, Erasmus University Rotterdam; 2018.
35. Versteegh M, Vermeulen KM, Evers SMAA, de Wit GA, Prenger R, Stolk EA. Dutch tariff for the five-level version of EQ-5D. Value Health. 2016;19(4):343–52.
36. Pullen LCE, Noortman WA, Triemstra L, de Jongh C, Rademaker FJ, Spijkerman R, et al. Prognostic value of [18F]FDG PET radiomics to detect peritoneal and distant metastases in locally advanced gastric cancer—a side study of the prospective multicentre PLASTIC study. Cancers (Basel). 2023 May 23;15(11):2874. https://doi.org/10.3390/cancers15112874.
37. Kawamura T, Kusakabe T, Sugino T, Watanabe K, Fukuda T, Nashimoto A, et al. Expression of glucose transporter-1 in human gastric carcinoma: association with tumor aggressiveness, metastasis, and patient survival. Cancer. 2001;92(3):634–41.
38. Zhao L, Pang Y, Luo Z, Fu K, Yang T, Zhao L, et al. Role of [^{68}Ga]Ga-DOTA-FAPI-04 PET/CT in the evaluation of peritoneal carcinomatosis and comparison with [^{18}F]-FDG PET/CT. Eur J Nucl Med Mol Imaging. 2021;48:1944–55.

Publisher's Note